MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3860 REAGENTS – LIMESTONE EQUIPMENT
DESIGN CALCULATION OF TANK FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The Purpose of this document is to analysis & design of **3860-REA-001 LIMESTONE MIXING TANK & 3860-REA-002 LIMESTONE DOSING TANK**. The dimensions and reinforcement are to be calculated. The report describes the methodology and the design parameters used in the design calculation. The analysis & design is carried out in accordance with project specifications , codes & standards.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah- Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in standard excel sheet. Stability check against Overturning & Sliding are done in the excel sheet. Base Pressure check has been done in the excel sheet. For strength calculation Bending moment & Shear Force values are calculate in the conventional method & designed in the standard excel sheet.

4. DESIGN REQUIREMENTS

4.1 Reference Codes & Books

Reference is made in this standard as indicated in STRUCTURAL ENGINEERING DESIGN CRITERIA Doc No. : MD-216-1000-EG-ST-CRT-1001-A03 , Structural Design Criteria - Process Building & Structures ITB Doc No. : MD-101-SMEM-EG-ST-CRT-1001 & Civil Design Criteria - General ITB Doc. No. MD-101-SMEM-EG-CV-CRT-1001.

The below mention codes & books are also used for this design calculation.

(i) SBC - 301
(ii) SBC - 304 CR - 2018
(iii) ACI 318M.
(iv) ACI SP-17M(09)

4.2 Materials Property

A. Density

4.2.A.(i) Density of Dry Soil : 1.9 T/m^3
4.2.A.(ii) Density of Concrete : 2.5 T/m^3
4.2.A.(iii) Density of Water : 1.0 T/m^3

B. Concrete Grade

4.2.B.(i) Foundation and Structural Members: 35 Mpa Strength of 28 days
4.2.B.(ii) Blinding Concrete below foundation bases: 15 Mpa Strength of 28 days

C. Reinforcement Grade

4.2.C.(i) Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
4.2.C.(ii) Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

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D. Grout Grade

4.2.D.(i) Grout shall be 50 mm cementitious of Strength 40 MPa.

4.3 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.4 Concrete Clear Cover

Concrete Exposure	Member	MinimumCove (mm)
Exposed to Soil	Foundation	50
	Wall	50
	Slab on Grade	20

5. LOAD

5.1 Primary Loads

3860-REA-001 LIMESTONE MIXING TANK & 3860-REA-002 LIMESTONE DOSING TANK foundation design the primary load values are taken from Drg. No. : MD-216-3860-EG-ME-PLN-1001 & MD-216-3860-EG-ME-PLN-1002 respectively.

1 DEAD LOAD OF EQUIPMENT (D)
2 EXTERNAL PLATFORM LOAD ($D_{platform}$)
3 OPERATIONAL WT. OF SLURRY ($F_{operating}$)
4 MAXIMUM WT. OF SLURRY (F_{max})
5 WIND LOAD (W)
7 SEISMIC LOAD (S)

5.2 Load Combinations for Stress Calculation

LOAD CASE	LOAD COMBINATION
101	$D + D_{platform} + F_{operation}$
102	$D + D_{platform} + F_{max}$
103	$D + D_{platform} + F_{operation} + W$
104	$D + D_{platform} + F_{max} + W$
105	$D + D_{platform} + F_{operation} + 0.7*E$
106	$D + D_{platform} + F_{max} + 0.7*E$
107	$0.6*(D + D_{platform}) + W$
108	$0.6*(D + D_{platform}) + 0.7*E$

5.3 Load Combinations for Strength Calculation

LOAD CASE	LOAD COMBINATION
201	$1.4*(D + D_{platform} + F_{operation})$
202	$1.4*(D + D_{platform} + F_{max})$
203	$1.2*(D + D_{platform}) + 0.8*W$
204	$1.2*(D + D_{platform}) + 1.6*W$
205	$1.2*(D + D_{platform}) + E$
206	$0.9*D + 1.6*W$
207	$0.9*D + E$

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6. SAFETY FACTOR AGAINST STABILITY

6.1 Overturning : For foundation design factor of safety against overturning is consider 2.0 for permanent load.
For foundation design factor of safety against overturning is consider 1.5 for temporary load.

6.2 Sliding : For foundation design a minimum factor of safety against sliding is consider 1.5

7. ALLOWABLE SOIL BEARING CAPACITY

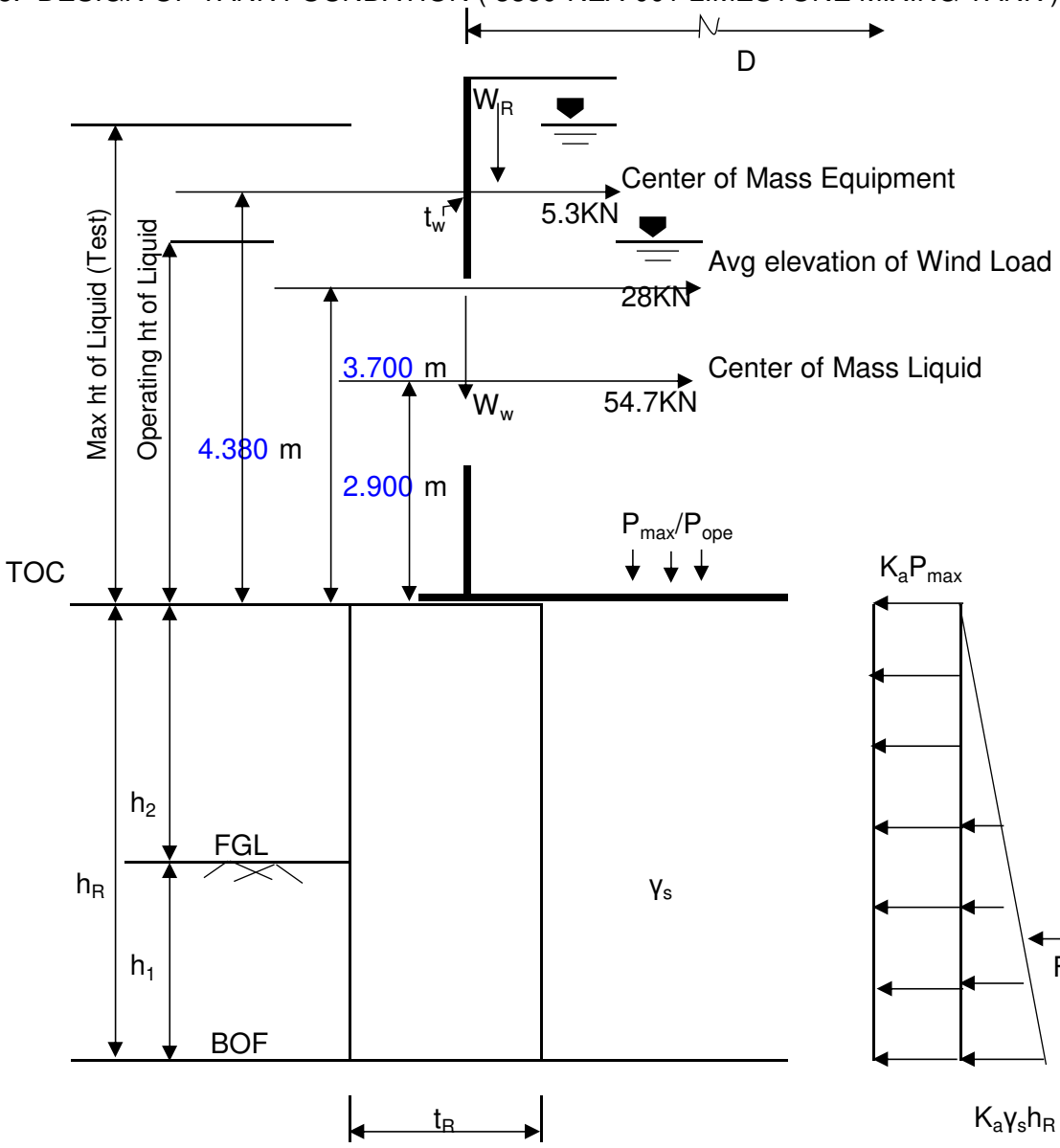
The net safe bearing capacity of soil for foundations design as obtained from page 39 of Doc. No. MD-216-0000-EM-GM-RPT-0003 is 670 KN/m2 at 1.2m depth below FGL
The neareast Bore Hole of 3850-REA-001 MILK OF LIME DOSING TANK is BH-12 to BH-15.
The net safe bearing capacity of soil as per approved L&T soil test report (Table-6 of Doc. No. MD-216-3000-EG-CV-RPT-1008-A02) is 1200 KN/m2 at 1.5m depth below FGL.

However for safe design purpose we have considered the Net Safe Bearing Capacity at founding level as **670 KN/m2**.

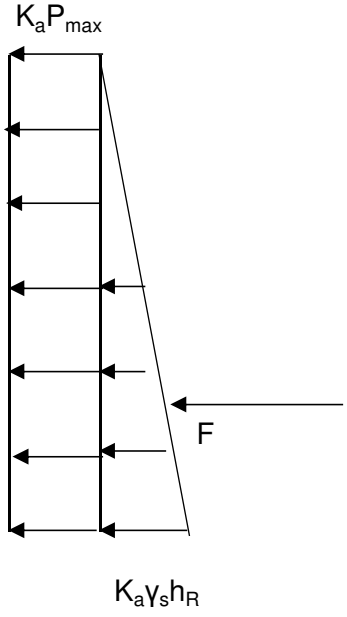
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8. DESIGN OF TANK FOUNDATION (3860-REA-001 LIMESTONE MIXING TANK)



Typical Section Through Tank



Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	5.012	m
Self weight of tank =	117	KN
Self weight of tank per m of circumference,W_T =	7.43	KN/m
External platform weight =	23.3	KN
Platform weight per m of circumference,W_P=	1.48	KN/m
Maximum weight of Slurry (F_{max}) =	1410	KN
Pressure at base of tank , P_{max} = (F_{max}) / A =	71.5	KN/m^2
Operating weight of Slurry ($F_{operating}$) =	1216	KN
Pressure at base of tank , $P_{operating}$ = ($F_{operating}$) / A =	61.6	KN/m^2

Foundation Material :

Concrete, f_c' =	35	MPa
Reinforcement, f_y =	420	MPa
Unit weight of concrete,γ_c =	24	KN/m^3

Soil Data :

Unit weight of soil,γ_s =	19	KN/m^3	(Natural sand)
Allowable bearing pressure, q_{all}=	670	KN/m^2	
Coefficient of active earth pressure,K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	3.2	m
FGL to Bottom of foundation ,h_1 =	1.5	m
FGL to Top of concrete ,h_2 =	1.7	m
Ringwall thickness, t_R =	550	mm
Inner dia of ringwall,D_i =	4.51	m
Outer dia of ringwall,D_o =	5.61	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	10.10	m^3

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Check of soil bearing pressure (Max weight of Slurry)

Wind force = 28 KN acting at a height of 6.900 m from BOF

M_{OT} = Overturning moment = 193.2 KN-m

M_{OT} / S_{xx} = 19.1 KN/m^2

Load Combination : 102. $D + D_{platform} + F_{max}$

$q_{RF} = (W_T + W_P)/t_R + P_{max} + h_R\gamma_c$ = 164 KN/m^2 **< 670KN/m2, O.K.**

Load Combination : 104. $D + D_{platform} + F_{max} \pm W$

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 184 KN/m^2 **< 670KN/m2, O.K.**

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 145 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : 107. $0.6*(D + D_{platform}) \pm W$

$q_{RE} = (W_T + W_P)/t_R + h_R\gamma_c$ = 93.0 KN/m^2

$q_{max} = 0.6*q_{RE} + M_{OT}/S_{xx}$ = 74.9 KN/m^2 **< 670KN/m2, O.K.**

$q_{min} = 0.6*q_{RE} - M_{OT}/S_{xx}$ = 36.68 KN/m^2 **No (-)ve pressure**

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 193.2 KN-m

M_{OR} = Restoring moment = (117+PI()*5.012*0.55*3.2*25)*5.012/2 = 2029 KN-m

FOS = M_{OR} / M_{OT} = 2029.38 / 193.2 = 10.50 **> 1.5, Hence O.K.**

Check for soil bearing pressure (Operating weight of slurry):

Seismic force at center of mass equipment = 117/(117+1216)*60 = 5.3 KN
acting at a height of 7.580 m from BOF

Seismic force at center of mass slurry = 1216/(117+1216)*60 = 54.7 KN
acting at a height of 6.100 m from BOF

M_{OT} = Overturning moment = 374 KN-m

M_{OT} / S_{xx} = 37 KN/m^2

Load Combination : 103. $D + D_{platform} + F_{operating}$

$q_{RF} = (W_T + W_P)/t_R + P_{operating} + h_R\gamma_c$ = 155 KN/m^2 **< 670KN/m2, O.K.**

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Load Combination : 107. $D + D_{platform} + F_{operating} \pm 0.7*E$

$q_{max} = q_{RF} + 0.7*M_{OT}/S_{xx}$ = 181 KN/m^2 **< 670KN/m2, O.K.**

$q_{min} = q_{RF} - 0.7*M_{OT}/S_{xx}$ = 129 KN/m^2

Load Combination : 111. $0.6*(D + D_{platform}) \pm 0.7*E$

$q_{max} = 0.6*(W_T + W_P)/t_R + 0.6*h_R\gamma_c + 0.7*M_{OT}/S_{xx}$ = 81.7 KN/m^2 **< 670KN/m2, O.K.**

$q_{max} = 0.6*(W_T + W_P)/t_R + 0.6*h_R\gamma_c - 0.7*M_{OT}/S_{xx}$ = 29.9 KN/m^2 **No (-)ve pressure**

Check for Overturning (Empty Tank): For Seismic Load

M_{OT} = Overturning moment = 373.8 KN-m

M_{OR} = Restoring moment = (117+PI()*5.012*0.55*3.2*25)*5.012/2 = 2029 KN-m

FOS = M_{OR} / M_{OT} = 2029.38 / 373.844 = 5.43 **> 1.5, Hence O.K.**

Ringwall Reinforcement :

Radial outward force on ringwall, $F= K_a h_R(P_{max} + \gamma_s h_R/2)$ = 108 KN/m

Hoop tension in ringwall, T = 1/2FD = 344 KN

Factored design tension, T_u= 481 KN FOS = 1.4

$A_{st,reqd} = T_u/\phi f_y$ = 1273 mm^2 ϕ = 0.9

$A_{st,min}$ = 0.002* $h_R t_R$ = 3520 mm^2

$A_{st,reqd}$ (per face of ring wall)= 1760 mm^2

Required : 12 dia bar @ 206 c/c (per face)

Provide : 12 dia bar @ 150 c/c (per face)

$A_{st,\ provided}$(per face) = 2413 mm^2 **O.K.**

Vertical Reinforcement :

$A_{st,min}$ = 0.0012* t_R = 660 mm^2/m

$A_{st,reqd}$ (per face of ring wall)= 330 mm^2

Required : 10 dia bar @ 238 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{st,\ provided}$(per face) = 393 mm^2 **O.K.**

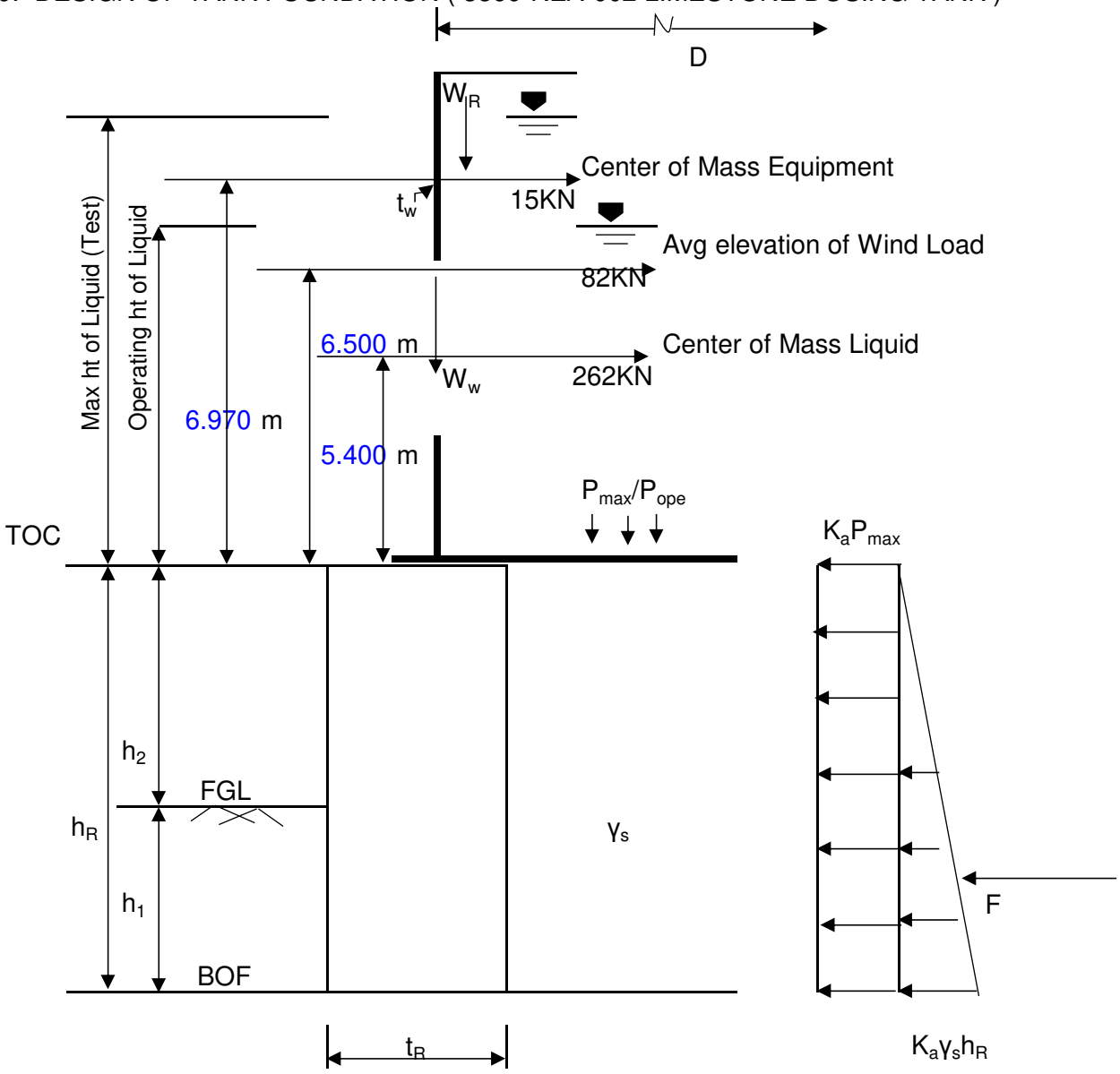
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9. DESIGN OF TANK FOUNDATION (3860-REA-002 LIMESTONE DOSING TANK)



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	8.012	m
Self weight of tank =	342	KN
Self weight of tank per m of circumference,W_T =	13.6	KN/m
External platform weight =	27.5	KN
Platform weight per m of circumference,W_P=	1.09	KN/m
Maximum weight of Slurry (F_{max}) =	6720	KN
Pressure at base of tank , P_{max} = (F_{max}) / A =	133	KN/m^2
Operating weight of Slurry ($F_{operating}$) =	5974	KN
Pressure at base of tank , $P_{operating}$ = ($F_{operating}$) / A =	118	KN/m^2

Foundation Material :

Concrete, f_c' =	35	MPa
Reinforcement, f_y =	420	MPa
Unit weight of concrete,γ_c =	24	KN/m^3

Soil Data :

Unit weight of soil,γ_s =	19	KN/m^3	(Natural sand)
Allowable bearing pressure, q_{all}=	670	KN/m^2	
Coefficient of active earth pressure,K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	3.2	m
FGL to Bottom of foundation ,h_1 =	1.5	m
FGL to Top of concrete ,h_2 =	1.7	m
Ringwall thickness, t_R =	550	mm
Inner dia of ringwall,D_i =	7.51	m
Outer dia of ringwall,D_o =	8.61	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	26.41	m^3

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MA'ADEN Saudi Arabian Mining Company		Outotec / LARSEN & TOUBRO
MD-216-3860-EG-CV-CAL-1433	Mansourah - Massarah Gold Project	Page: of
900707_GAU01_AA23_CED01_ 38601_OU100690359_R2	DESIGN CALCULATION REAGENTS – LIMESTONE EQUIPMENT DESIGN CALCULATION OF TANK FOUNDATION	Date: 30.11.2023
O19045_450_C_K_24169_001_R2	AS BUILT	Rev: D01

Check of soil bearing pressure (Max weight of Slurry)

Wind force = 82 KN acting at a height of 9.700 m from BOF

M_{OT} = Overturning moment = 795.4 KN-m

M_{OT} / S_{xx} = 30.1 KN/m^2

Load Combination : 102. $D + D_{platform} + F_{max}$

$q_{RF} = (W_T + W_P)/t_R + P_{max} + h_R\gamma_c$ = 237 KN/m^2 **< 670KN/m2, O.K.**

Load Combination : 104. $D + D_{platform} + F_{max} \pm W$

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 267 KN/m^2 **< 670KN/m2, O.K.**

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 207 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : 107. $0.6*(D + D_{platform}) \pm W$

$q_{RE} = (W_T + W_P)/t_R + h_R\gamma_c$ = 103.5 KN/m^2

$q_{max} = 0.6*q_{RE} + M_{OT}/S_{xx}$ = 92.2 KN/m^2 **< 670KN/m2, O.K.**

$q_{min} = 0.6*q_{RE} - M_{OT}/S_{xx}$ = 31.97 KN/m^2 **No (-)ve pressure**

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 795.4 KN-m

M_{OR} = Restoring moment = (342+PI()*8.012*0.55*3.2*25)*8.012/2 = 5807 KN-m

FOS = M_{OR} / M_{OT} = 5806.69 / 795.4 = 7.30 **> 1.5, Hence O.K.**

Check for soil bearing pressure (Operating weight of slurry):

Seismic force at center of mass equipment = 342/(342+5974)*277 = 15 KN
acting at a height of 10.170 m from BOF

Seismic force at center of mass slurry = 5974/(342+5974)*277 = 262.0 KN
acting at a height of 8.600 m from BOF

M_{OT} = Overturning moment = 2406 KN-m

M_{OT} / S_{xx} = 91.1 KN/m^2

Load Combination : 103. $D + D_{platform} + F_{operating}$

$q_{RF} = (W_T + W_P)/t_R + P_{operating} + h_R\gamma_c$ = 222 KN/m^2 **< 670KN/m2, O.K.**

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MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec / LARSEN & TOUBRO
MD-216-3860-EG-CV-CAL-1433	Mansourah - Massarah Gold Project	Page: of
900707_GAU01_AA23_CED01_ 38601_OU100690359_R2	DESIGN CALCULATION REAGENTS – LIMESTONE EQUIPMENT DESIGN CALCULATION OF TANK FOUNDATION	Date: 30.11.2023
O19045_450_C_K_24169_001_R2	AS BUILT	Rev: D01

Load Combination : 107. $D + D_{platform} + F_{operating} \pm 0.7*E$

$q_{max} = q_{RF} + 0.7*M_{OT}/S_{xx}$ = 286 KN/m^2 **< 670KN/m2, O.K.**

$q_{min} = q_{RF} - 0.7*M_{OT}/S_{xx}$ = 158 KN/m^2

Load Combination : 111. $0.6*(D + D_{platform}) \pm 0.7*E$

$q_{max} = 0.6*(W_T + W_P)/t_R + 0.6*h_R\gamma_c + 0.7*M_{OT}/S_{xx}$ = 126 KN/m^2 **< 670KN/m2, O.K.**

$q_{max} = 0.6*(W_T + W_P)/t_R + 0.6*h_R\gamma_c - 0.7*M_{OT}/S_{xx}$ = -1.68 KN/m^2

Check for Overturning (Empty Tank): For Seismic Load

M_{OT} = Overturning moment = 2405.8 KN-m

M_{OR} = Restoring moment = (342+PI()*8.012*0.55*3.2*25)*8.012/2 = 5807 KN-m

FOS = M_{OR} / M_{OT} = 5806.69 / 2405.75 = 2.41 **> 1.5, Hence O.K.**

Ringwall Reinforcement :

Radial outward force on ringwall, $F= K_a h_R(P_{max} + \gamma_s h_R/2)$ = 173 KN/m

Hoop tension in ringwall, T = 1/2FD = 692 KN

Factored design tension, T_u= 969 KN FOS = 1.4

$A_{st,reqd} = T_u/\phi f_y$ = 2565 mm^2 ϕ = 0.9

$A_{st,min}$ = 0.002* $h_R t_R$ = 3520 mm^2

$A_{st,reqd}$ (per face of ring wall)= 1760 mm^2

Required : 12 dia bar @ 206 c/c (per face)

Provide : 12 dia bar @ 150 c/c (per face)

$A_{st,\ provided}$(per face) = 2413 mm^2 **O.K.**

Vertical Reinforcement :

$A_{st,min}$ = 0.0012* t_R = 660 mm^2/m

$A_{st,reqd}$ (per face of ring wall)= 330 mm^2

Required : 10 dia bar @ 238 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{st,\ provided}$(per face) = 393 mm^2 **O.K.**

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ATTACHMENT - 1 : Mechanical Input Drawing.

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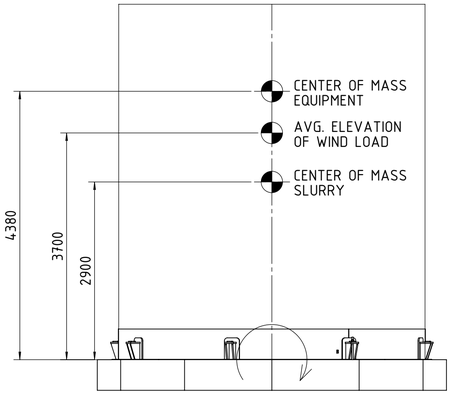
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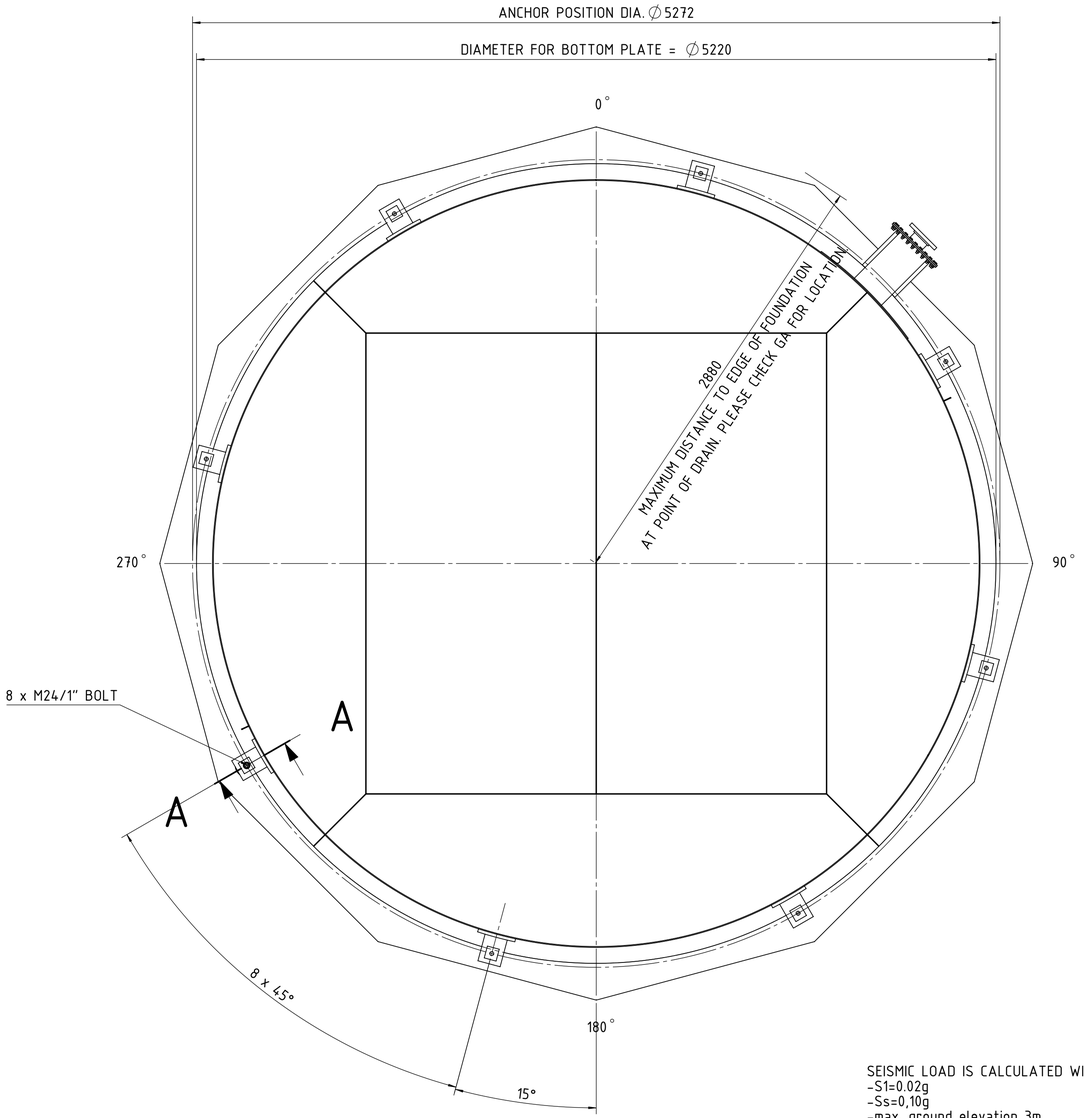
DRAWING NUMBER	REV
MD-216-3860-EG-ME-PLN-1001	A02

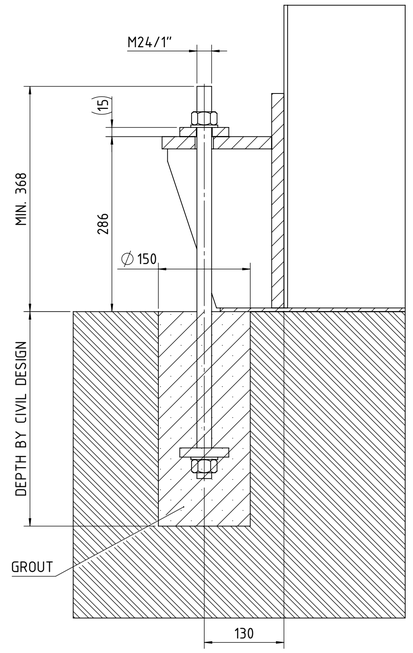
LOADING NOTES:	
1.	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2.	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
3.	REPORTED SEISMIC LOADS ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC 301 - 2007. HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNERACCORDING SBC 301 - 2007.
4.	REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC 301 - 2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE SHELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT

2



SCALE 1 : 50





SECTION A-A
SCALE 1 : 5

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. ground elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

LIMESTONE MIXING TANK	
DEAD WEIGHT OF EQUIPMENT	11700 kg
OPERATIONAL WEIGHT OF SLURRY	121600 kg
MAXIMUM WEIGHT OF SLURRY	141000 kg
EXTERNAL PLATFORM LOAD	2330 kg
NORMAL OPERATING SPEED	55,7 rpm
WIND LOAD	28 kN
SEISMIC LOAD	60 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	181 kNm

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3860-EG-ME-GAD-1001	GENERAL ARRANGEMENT DRAWING								
MD-216-3860-EG-ME-DAT-1002	EQUIPMENT DATASHEET								
MD-216-3860-EG-PR-PID-1001	P&I DIAGRAM	A02	03.03.2020	ISSUED FOR APPROVAL	VP	TM	JR	GR	
MD-216-3860-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	15.01.2020	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

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LARSEN & TOUBRO DRAWING NUMBER	019045-450-M-D-03215-002	REVISION 2
OUTOTEC DRAWING NUMBER	900707 GAU01 AA23 MLD15 38601	DOCUMENT ID 0U601269873 REVISION 2

DRAWN	vilpoy	27.02.2020
CHECKED	tuoman	03.03.2020
ENGINEER	juhris	03.03.2020
APPROVED	GeoRie	03.03.2020
EMPLOYER		
SCALE	1:20	A1

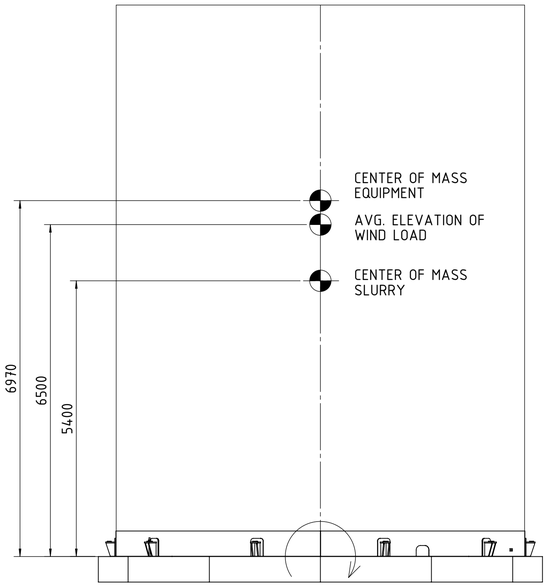
MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
LIMESTONE PREPARATION
3860-REA-001 LIMESTONE MIXING TANK

DRAWING NUMBER MD-216-3860-EG-ME-PLN-1001 REV A02

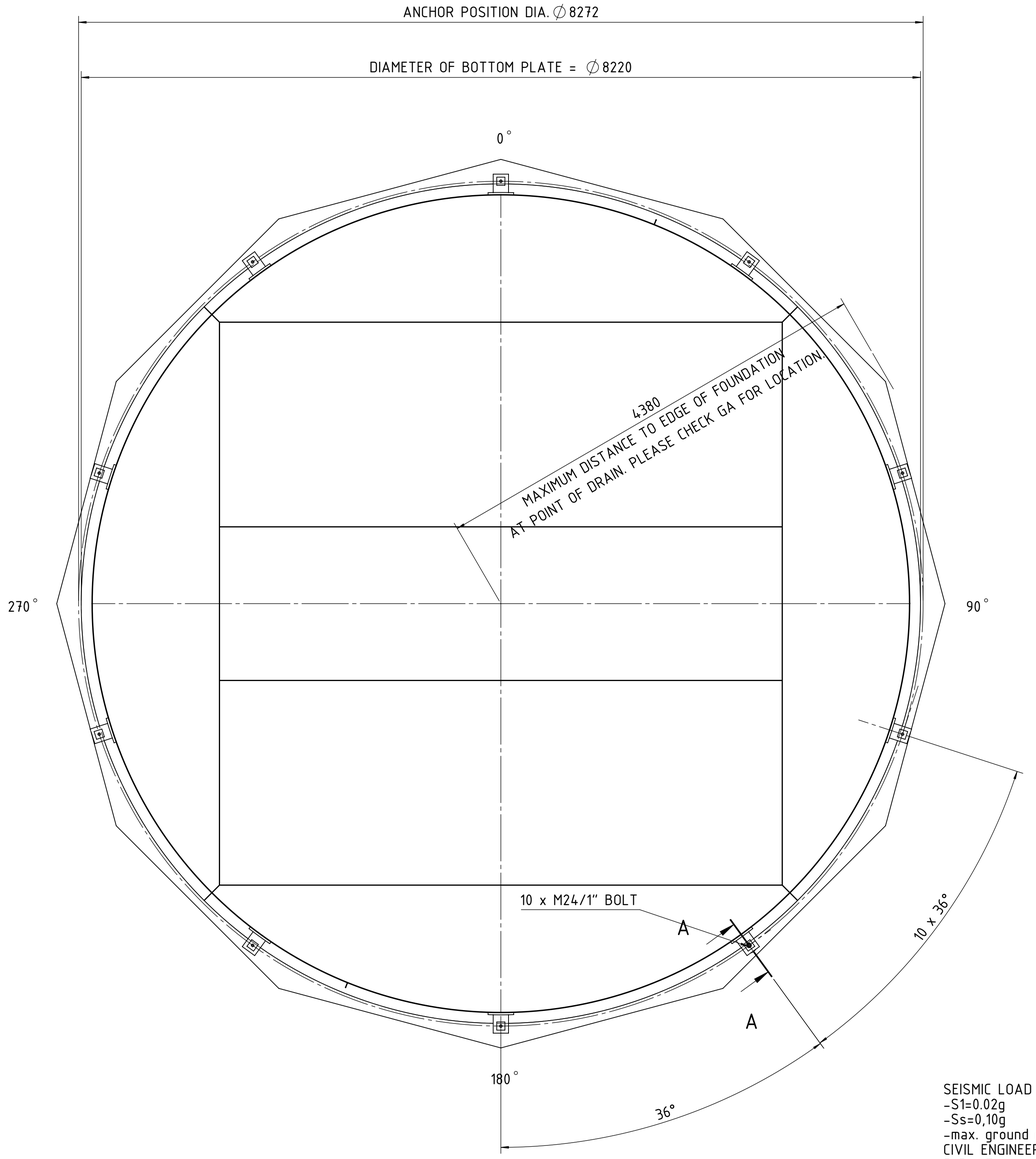
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MD-216-3860-EG-ME-PLN-1002	A02

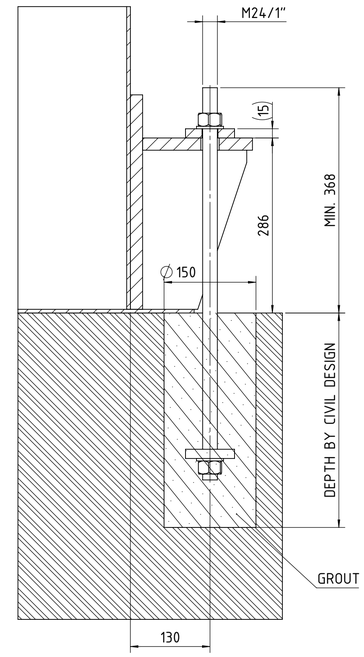
LOADING NOTES:	
1.	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
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3.	REPORTED SEISMIC LOADS ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC 301 - 2007. HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNERACCORDING SBC 301 - 2007.
4.	REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC 301 - 2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN
FOUNDATION AND SUPPORT TOLERANCES	
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1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT

2



SCALE 1 : 60





SECTION A-A
SCALE 1 : 5

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. ground elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

LIMESTONE DOSING TANK	
DEAD WEIGHT OF EQUIPMENT	34200 kg
OPERATIONAL WEIGHT OF SLURRY	597400 kg
MAXIMUM WEIGHT OF SLURRY	672000 kg
EXTERNAL PLATFORM LOAD	2750 kg
NORMAL OPERATING SPEED	32 rpm
WIND LOAD	82 kN
SEISMIC LOAD	277 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	1517 kNm

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3860-EG-ME-GAD-1002	GENERAL ARRANGEMENT DRAWING								
MD-216-3860-EG-ME-DAT-1003	EQUIPMENT DATASHEET								
MD-216-3860-EG-PR-PID-1002	P&I DIAGRAM	A02	03.03.2020	ISSUED FOR APPROVAL	VP	TM	JR	GR	
MD-216-3860-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	15.01.2020	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

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OUTOTEC DRAWING NUMBER: 900707 GAU01 AA23 MLD15 38602 — DOCUMENT ID: OU601269783 — REVISION: 2

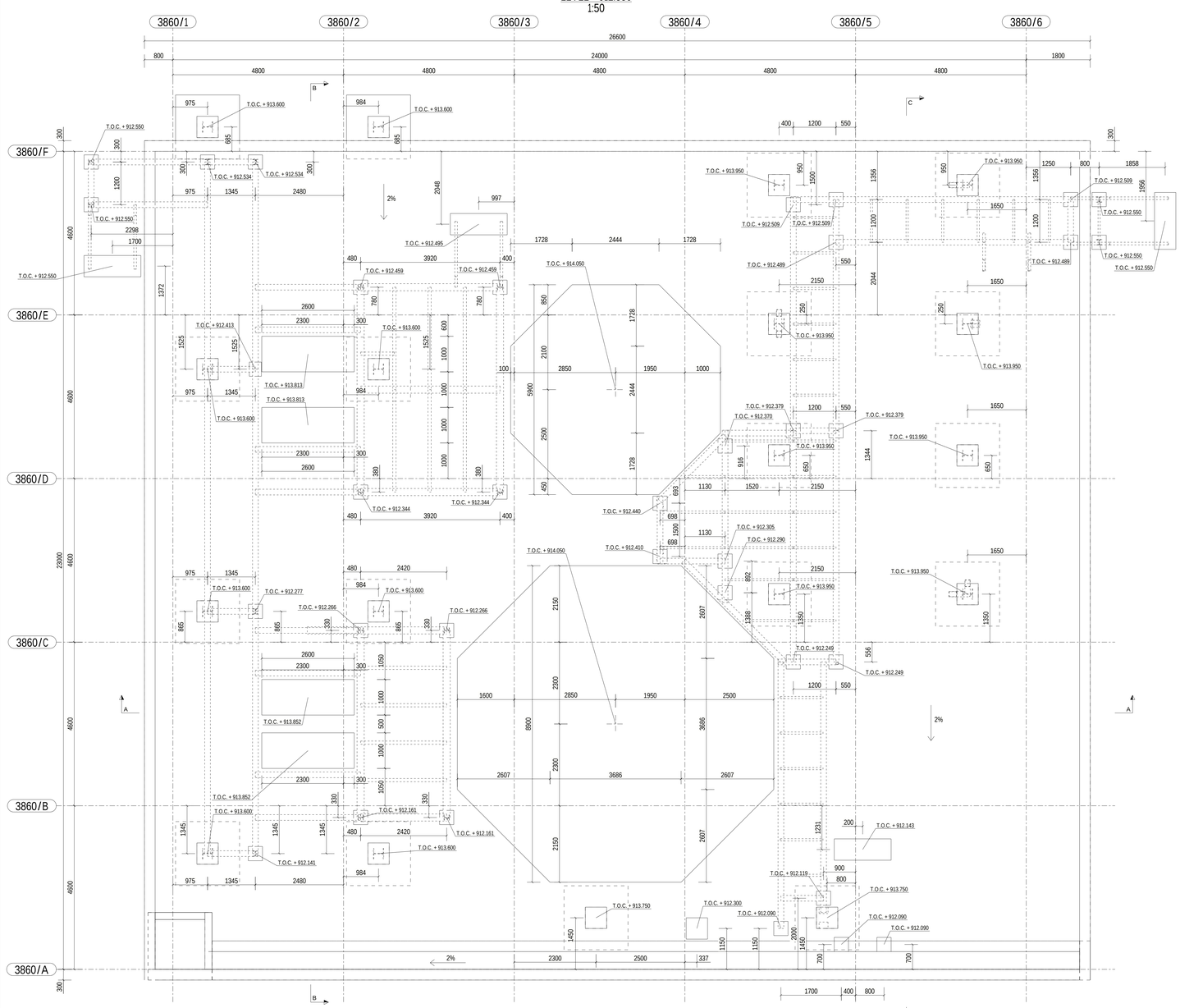
DRAWN	vilpoy	27.02.2020
CHECKED	tuoman	03.03.2020
ENGINEER	juhris	03.03.2020
APPROVED	GeoRie	03.03.2020
EMPLOYER		
SCALE	1:30	A1

MANSOURAH-MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
LIMESTONE PREPARATION
3860-REA-002 LIMESTONE DOSING TANK

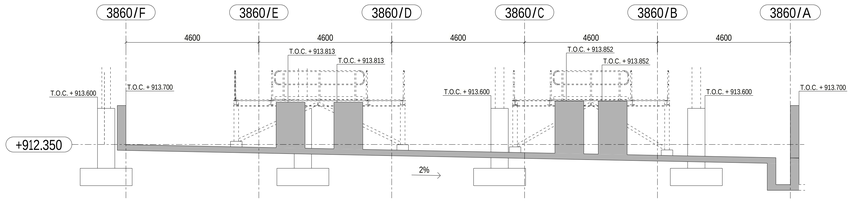
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DRAWING NUMBER MD-216-3860-EG-CV-GAD-1001 REV A01

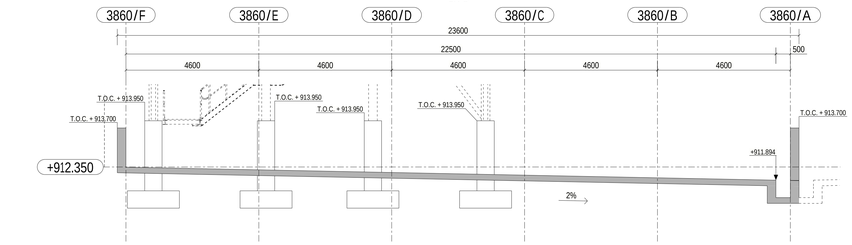
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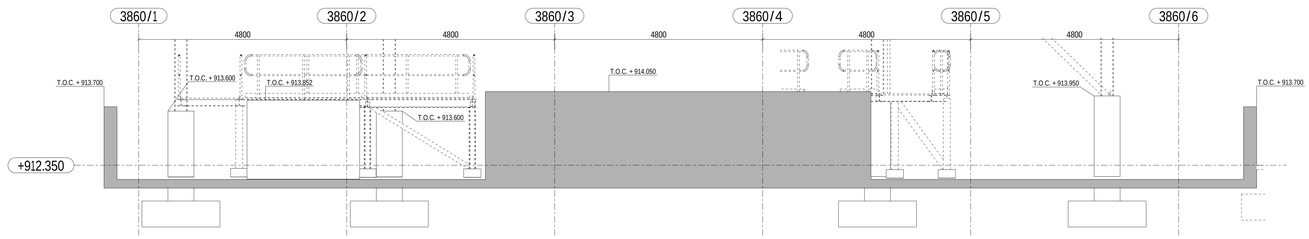
B - B
1:75



C - C
1:75



A - A
1:50



THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3850 = RL(+)912.350 m

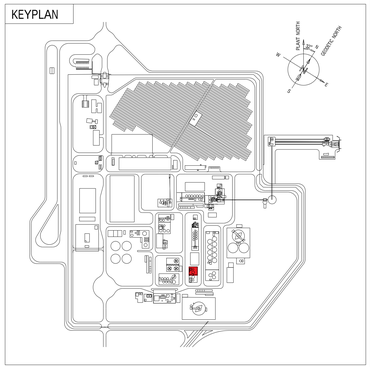
Document Number	MD-216-3860-EG-CV-GAD-1001
Revision	A01
Approved	Mr Doron Golan - Bechtel Global Mar 4, 2020, 4:59 PM GMT+2:00

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- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ACID RESISTANT COATING
- CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION
MD-216-1000-EG-ST-CRT-0001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3860-EG-ST-GAD-1001	PLAN VIEW 3860/4-6/C-F, 3D VIEW AND SECTIONS			
MD-216-3850-EG-ME-GAD-1008	SECTION B-B	MD-216-3850-EG-ME-GAD-1009	SECTION C-C			
MD-216-3850-EG-ME-GAD-1006	PLAN VIEW	MD-216-3850-EG-ME-GAD-1007	SECTION A-A			
MD-216-3850-EG-ME-GAD-1004	PLAN AT EL. +912.350	MD-216-3850-EG-ME-GAD-1005	PLAN AT EL. +919.990	A01	07.02.2020	ISSUED FOR APPROVAL

	DRN	CHK	ENG	APP	EMP
REVISION DESCRIPTION	Kir.Jok	MikLai	LuiPiz	SuoCha	

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LARSEN & TOUBRO DRAWING NUMBER — REVISION 0

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MANSOURAH - MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
LIMESTONE
PLAN AT EL. +912.350 AND SECTIONS

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Trim Structure